

Jeffrey "Jack" Hannon Jr. — Resume

Jack Hannon

Linux / HPC Systems Administration · Automation · Diagnostics

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Summary

Linux-focused systems administrator working in a university HPC environment. Day-to-day work includes Slurm-driven cluster operations, hardware triage, firmware/BIOS coordination with vendors, and automation with shell and Python. I build tools to reduce repetitive operational pain and make cluster behavior more visible.

Skills

Systems & HPC: Linux (Debian, Rocky, Ubuntu), Slurm, Lustre, OFED, BIOS/firmware coordination

Automation & Scripting: Bash/Zsh, Python, basic C, Ansible, Git

Tooling & Platforms: ReFrame, Kubernetes (lab environments), QEMU/KVM, GitHub/Gitea

Other: Documentation, incident triage, vendor communication, internal tooling

Experience

Computer Specialist I — High-Performance Computing Center

Mississippi State University, Starkville, MS · 2022–Present

- Maintain and troubleshoot compute nodes across clusters ranging from 200 to 1,800 nodes, including drain handling, node reboots, hardware diagnosis, and vendor RMA coordination.
- Coordinate firmware, BIOS, and OFED upgrades with Dell and internal teams; run test jobs and document behavioral changes.
- Write and maintain diagnostic scripts (bash, Python) to surface node state, drain reasons, file system usage, and other health signals.
- Support researchers and internal staff with job failures, environment issues, and general HPC usage questions.

Research Computing Administrator (Freelance / Contract)

Remote · 2023–2024

- Assisted small research groups with Linux server setup, configuration, and basic automation for compute/storage.
- Designed simple, reproducible configurations for lab environments that could be rebuilt

from documentation and scripts.

- Operated independently with full responsibility for uptime, environment consistency, and recovery.

GCP Data Engineer — Tata Consultancy Services

Remote · 2021–2022

- Short-term contract role; gained practical exposure to GCP compute/storage and Hadoop-based data pipelines.

Senior Intern — Center for Cyber Innovation

Mississippi State University, Starkville, MS · 2020–2021

- Provided technical support for end users and internal staff, including OS troubleshooting, basic scripting, and system maintenance.
 - Contributed to basic documentation and small internal tools that reduced repeat support requests.
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Selected Projects

bad — HPC Administration Terminal UI

Tech: Python, curses, PTY, SLURM, NHC, iDRAC/RACADM, pytest

- Built a persistent interactive TUI replacing ad-hoc shell workflows for day-to-day cluster administration across large node sets
- Integrates SLURM, NHC, and Dell iDRAC RACADM into a single stateful session with live output capture, drainlist diff tracking, and cached sudo credentials
- ~8,500 lines across ~90 modules with 100+ pytest files including PTY simulation for integration-level testing

Home Network Lab — vplan & stepfam

Tech: Linux, WireGuard/NordVPN, iptables, Pi-hole, step-ca, Docker, Caddy, bash, Python

- Designed and maintain a multi-segment home network with a dual-NIC Debian Linux router providing LAN-wide VPN tunneling with selective per-device bypass via iptables mangle and policy routing tables
- Built vplan: a device-management toolkit generating Pi-hole DHCP reservations and DNS records idempotently from a single source-of-truth config, with full DNS postcheck validation
- Built stepfam: a two-part PKI toolkit managing a private ACME CA (step-ca in Docker) with inventory-driven cert issuance, SSH-based deployment, trust-chain distribution, and CA rotation — integrated with Caddy for automatic TLS on self-hosted services

netcon — iOS Network Coverage Mapper

Tech: Swift, SwiftUI, CoreLocation, CoreMotion, CoreData, MapKit, CoreGraphics

- Built a self-contained iOS app that passively measures Wi-Fi and cellular quality (latency, loss, throughput) as the user moves and renders results as a zoomable heatmap overlay

- Implemented a custom MapKit tile rasteriser with manual Web-Mercator projection, zoom-adaptive power-of-2 grid snapping, and confidence-based opacity blending
- Zero third-party dependencies; covers background location scheduling, Core Data persistence, and multi-format export (zip, CSV, GeoJSON)

myvm — Apple Silicon VM Manager

Tech: Python, PySide6, QEMU, QMP, HVF, cloud-init

- Desktop GUI for full VM lifecycle management on Apple Silicon using QEMU with HVF acceleration and macosvm for macOS guests
- Drives QEMU over QMP Unix socket for graceful ACPI shutdown; auto-manages SSH config blocks so VMs are immediately reachable by name after first boot
- Generates cloud-init NoCloud seed ISOs with Jinja-templated user-data for reproducible guest provisioning

Certifications/Awards

CompTIA A+

Awarded: 11/2025

Education

B.S. in Computer Science

Mississippi State University · 2021